

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12413046
Kind Code	B2
Date of Patent	September 09, 2025
Inventor(s)	Chen; Lih-Ren et al.

Photonic crystal surface emitting laser device

Abstract

A photonic crystal surface emitting laser device including at least one photonic crystal surface emitting laser unit is provided. The photonic crystal surface emitting laser unit includes a light-emitting layer, a photonic crystal layer, a doped semiconductor layer, and a diffractive grating. The light-emitting layer is configured to emit a light beam. The photonic crystal layer is disposed on one side of the light-emitting layer. The doped layer is disposed on another side of the light-emitting layer. The diffractive grating is disposed on the photonic crystal layer or the doped semiconductor layer.

Inventors: Chen; Lih-Ren (Hsinchu, TW), Hong; Kuo-Bin (Hsinchu, TW), Lu; Tien-Chang (Hsinchu, TW), Lin; Chien-Hung (Hsinchu, TW), Chen; Hsiu-Ling (Hsinchu, TW), Huang; Kuan-Chih (Hsinchu, TW)

Applicant: Phosertek Corporation (Hsinchu, TW)

Family ID: 83449188

Assignee: Phosertek Corporation (Hsinchu, TW)

Appl. No.: 17/709442

Filed: March 31, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20220320827 A1	Oct. 06, 2022

Foreign Application Priority Data

TW	110116735	May. 10, 2021
----	-----------	---------------

Related U.S. Application Data

Publication Classification**Int. Cl.:** **H01S5/11** (20210101); **H01S5/042** (20060101); **H01S5/183** (20060101)**U.S. Cl.:****CPC** **H01S5/11** (20210101); **H01S5/04256** (20190801); **H01S5/183** (20130101);**Field of Classification Search****CPC:** H01S (5/11); H01S (5/04256); H01S (5/183); H01S (5/0207); H01S (5/02255); H01S (5/42); H01S (5/185); H01S (5/04254); H01S (5/04257)

References Cited**U.S. PATENT DOCUMENTS**

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
10923633	12/2020	Ho et al.	N/A	N/A
2002/0106160	12/2001	Cox	372/50.1	H01S 5/18386
2019/0252855	12/2018	Lu	N/A	H01S 5/2059
2023/0198224	12/2022	Kurosaka	372/38.05	H01S 5/42

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
110247302	12/2018	CN	N/A
112673535	12/2020	CN	N/A
6213915	12/2016	JP	N/A
I381602	12/2012	TW	N/A
201917964	12/2018	TW	N/A
201935787	12/2018	TW	N/A
M615662	12/2020	TW	N/A

OTHER PUBLICATIONS

Lih-Ren Chen et al., "Integration of multiple diffractive elements on photonic crystal surface emitting lasers for beam manipulation", CLEO 2020, May 2020, Grace Period Disclosure, pp. 1-2. cited by applicant

Lih-Ren Chen et al., "Direct Integration of Beam Steering Structure on Photonic Crystal Surface Emitting Lasers", 2020 Opto-Electronics and Communications Conference (OECC), Oct. 2020, Grace Period Disclosure, pp. 1-3. cited by applicant

Lih-Ren Chen et al., "Vertically integrated diffractive gratings on photonic crystal surface emitting lasers", Scientific reports, Dec. 2021, Grace Period Disclosure, pp. 1-7. cited by applicant

Ryoichi Sakata et al., "Dually modulated photonic crystals enabling high-power high-beam-quality two-dimensional beam scanning lasers", Nature Communications, Jul. 2020, pp. 1-10. cited by applicant

Han-Lun Chiu et al., "Photonic Crystal Surface Emitting Lasers with Naturally Formed Periodic ITO Structures", ACS Photonics, Jan. 31, 2019, pp. 684-690. cited by applicant

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This application claims the priority benefit of U.S. provisional application Ser. No. 63/169,255 filed on Apr. 1, 2021, and Taiwan application serial no. 110116735 filed on May 10, 2021. The entirety of the above-mentioned patent applications are hereby incorporated by reference herein and made a part of this specification.

BACKGROUND

Technical Field

(1) The disclosure relates to a laser device, and in particular relates to a photonic crystal surface emitting laser device.

Description of Related Art

(2) An electroluminescent photonic crystal surface emitting laser may realize single mode output, narrow spectrum wavelength linewidth, and small exiting light divergence angle. The main structure of the electroluminescent photonic crystal surface emitting laser includes a bottom cladding layer, a light-emitting layer and a photonic crystal layer, where the light-emitting layer is located between the bottom cladding layer and the photonic crystal layer. Under this structure and operating mechanism, a laser light is emitted to the outside from at least one of the directions from a top portion or a bottom portion.

(3) In combination with a micro-electro mechanical system (MEMS) optical device or a mechanically oscillating optical device such that the photonic crystal surface emitting laser generates dynamic deflection, a scanning effect may be achieved. However, both the micro-electro mechanical system and the mechanically oscillating optical device achieve a scanning effect by the mechanism of oscillation, thus issues such as the scanning speed being limited by the mechanical oscillating speed, and the reliability and durability of the devices being insufficient may arise.

SUMMARY

(4) The disclosure provides a photonic crystal surface emitting laser device, which may determine an exiting light deflection direction of a laser beam by using a simple structure.

(5) An embodiment of the disclosure provides a photonic crystal surface emitting laser device, including at least one photonic crystal surface emitting laser unit. The photonic crystal surface emitting laser unit includes a light-emitting layer, a photonic crystal layer, a doped semiconductor layer, and a diffractive grating. The light-emitting layer is configured to emit a light beam, and the photonic crystal layer is disposed on one side of the light-emitting layer. The doped semiconductor layer is disposed on another side of the light-emitting layer, and the diffractive grating is disposed on the photonic crystal layer or the doped semiconductor layer.

(6) In the photonic crystal surface emitting laser device of the embodiment of the disclosure, since the diffractive grating disposed on the photonic crystal layer or the doped semiconductor layer is adopted to diffract the laser beam diffracted by the photonic crystal layer, therefore the exiting light deflection direction of the laser beam may be determined by using a simple structure.

Description

BRIEF DESCRIPTION OF THE DRAWING

- (1) FIG. 1A is a perspective schematic view of a photonic crystal surface emitting laser device of an embodiment of the disclosure.
- (2) FIG. 1B is a top schematic view of the photonic crystal surface emitting laser device of FIG. 1A.
- (3) FIG. 1C is a cross-sectional schematic view of the photonic crystal surface emitting laser device of FIG. 1B along a line I-I.
- (4) FIG. 1D and FIG. 1E are respective top schematic views of the photonic crystal surface emitting laser device of another two embodiments of the disclosure.
- (5) FIG. 2A is a cross-sectional schematic view of a photonic crystal surface emitting laser device of another embodiment of the disclosure.
- (6) FIG. 2B is a simplified perspective schematic view of a photonic crystal layer, a diffractive grating surface of a transparent conductive layer, and a diffractive grating in FIG. 2A.
- (7) FIG. 3 is a cross-sectional schematic view of a photonic crystal surface emitting laser device of yet another embodiment of the disclosure.
- (8) FIG. 4 is a cross-sectional schematic view of a photonic crystal surface emitting laser device of another embodiment of the disclosure.
- (9) FIG. 5 is a cross-sectional schematic view of a photonic crystal surface emitting laser device of yet another embodiment of the disclosure.
- (10) FIG. 6 is a cross-sectional schematic view of a photonic crystal surface emitting laser device of yet another embodiment of the disclosure.
- (11) FIG. 7 is a cross-sectional schematic view of a photonic crystal surface emitting laser device of yet another embodiment of the disclosure.
- (12) FIG. 8 is a cross-sectional schematic view of a photonic crystal surface emitting laser device of another embodiment of the disclosure.
- (13) FIG. 9 is a cross-sectional schematic view of a photonic crystal surface emitting laser device of yet another embodiment of the disclosure.
- (14) FIG. 10 is a cross-sectional schematic view of a photonic crystal surface emitting laser device of yet another embodiment of the disclosure.
- (15) FIG. 11 is a cross-sectional schematic view of a photonic crystal surface emitting laser device of yet another embodiment of the disclosure.
- (16) FIG. 12 is a cross-sectional schematic view of a photonic crystal surface emitting laser device of yet another embodiment of the disclosure.

DETAILED DESCRIPTION OF DISCLOSED EMBODIMENTS

- (17) FIG. 1A is a perspective schematic view of a photonic crystal surface emitting laser device of an embodiment of the disclosure, FIG. 1B is a top schematic view of the photonic crystal surface emitting laser device of FIG. 1A, and FIG. 1C is a cross-sectional schematic view of the photonic crystal surface emitting laser device of FIG. 1B along a line I-I. In FIG. 1A, a top portion of the photonic crystal surface emitting laser device is partially cut open, to show a portion of the internal structure. Referring to FIG. 1A, FIG. 1B, and FIG. 1C, a photonic crystal surface emitting laser device **200** of this embodiment includes at least one photonic crystal surface emitting laser unit **100** (FIG. 1A to FIG. 1C are illustrated with a photonic crystal surface emitting laser unit **100** as an example). The photonic crystal surface emitting laser unit **100** includes a light-emitting layer **120**, a photonic crystal layer **130**, a doped semiconductor layer **110**, and a diffractive grating **300**. The light-emitting layer **120** is configured to emit a light beam **122**. In this embodiment, the light-emitting layer **120** includes a quantum well layer, a multiple quantum well layer, or a quantum dot layer, and the light beam **122** emitted by the light-emitting layer **120** may be an infrared light, a visible light or an ultraviolet light.
- (18) The photonic crystal layer **130** is disposed on one side of the light-emitting layer **120**, and the

doped semiconductor layer **110** is disposed on another side of the light-emitting layer **120**. In this embodiment, the photonic crystal layer **130** is a P-type semiconductor layer, and the doped semiconductor layer **110** is an N-type semiconductor layer. However, in other embodiments, the photonic crystal layer **130** may be an N-type semiconductor layer, and the doped semiconductor layer **110** may be a P-type semiconductor layer. In addition, the diffractive grating **300** is disposed on the photonic crystal layer **130** or the doped semiconductor layer **110**, and FIG. **1A** and FIG. **1C** are examples where the diffractive grating **300** is disposed on the photonic crystal layer **130**.

(19) In this embodiment, the photonic crystal surface emitting laser device **200** further includes a first electrode **140** electrically connected to the doped semiconductor layer **110**. The photonic crystal surface emitting laser unit **100** further includes a second electrode **150** disposed on the photonic crystal layer **130** and electrically connected to the photonic crystal layer **130**. In this embodiment, by applying a forward bias voltage to the second electrode **150** and the first electrode **140**, the electron holes in the photonic crystal layer **130** and the electrons in the doped semiconductor layer **110** may be mobilized into the light-emitting layer **120**, and recombined in the light-emitting layer **120** to generate the light beam **122**.

(20) The light beam **122** emitted by the light-emitting layer **120** generates a resonance in a horizontal direction in FIG. **1C** in the photonic crystal layer **130**, and the photonic crystal layer **130** may include a second-order grating structure, which directs the light beam **122** to a vertical direction in FIG. **1C**, transmitting the light beam **122** toward the top and bottom of FIG. **1C**. That is, the light-emitting layer **120** is a gain medium that emits the light beam **122** and provides an optical gain. After the light beam **122** enters the photonic crystal layer **130** and generates a diffraction resonance, a laser beam radiating in the vertical direction is generated. In addition, the structure of the second-order grating may be a structure with a two-dimensional hole array. In other embodiments, the photonic crystal layer **130** may also be a grating structure with a third or higher order, and in addition, the photonic crystal layer **130** may also be a one-dimensional grating structure. After the light beam **122** is diffracted by the photonic crystal layer **130** and the diffractive grating **300**, the light beam **122** is emitted from the photonic crystal surface emitting laser unit **100**. In other embodiments, the photonic crystal layer **130** may also include paired up circular or rectangular holes in a configuration of a two-dimensional array, a triangular hole array, or various other possible photonic crystal structures.

(21) In the photonic crystal surface emitting laser device **200** of this embodiment, the diffractive grating **300** disposed on the photonic crystal layer **130** or the doped semiconductor layer **110** is adopted to diffract the light beam **122** diffracted by the photonic crystal layer **130** (i.e., the laser beam). Therefore, the exiting deflection direction of the light beam **122** (i.e., the laser beam) may be determined by using a simple structure.

(22) In this embodiment, a portion of the second electrode **150** is the diffractive grating **300**. For example, as shown in FIG. **1A** and FIG. **1C**, periodically arranged elongated slits **152** are formed at a center portion of the second electrode **150** to form the diffractive grating **300**. However, in other embodiments, there may be an opening in the center of the second electrode **150**, and a diffractive grating **300** formed of a material different from that of the second electrode **150** is disposed in the opening. The material of the diffractive grating **300** may be a metal, a semiconductor, or an oxide.

(23) In this embodiment, the photonic crystal surface emitting laser unit **100** further includes a transparent conductive layer **170** disposed between the photonic crystal layer **130** and the second electrode **150** to electrically connect the photonic crystal layer **130** and the second electrode **150**. In this embodiment, the photonic crystal surface emitting laser device **200** further includes a substrate **160** disposed between the doped semiconductor layer **110** and the first electrode **140**, and electrically connected to the doped semiconductor layer **110** and the first electrode **140**, in addition, the substrate **160** has conductivity.

(24) In this embodiment, the photonic crystal layer **130** includes a cladding layer **132**, a graded index layer (GRIN layer) **136**, and an ohmic contact layer **134**. The cladding layer **132** is disposed

on the light-emitting layer **120**, and the graded index layer **136** is disposed on the cladding layer **132**. The ohmic contact layer **134** is disposed between the graded index layer **136** and the transparent conductive layer **170**, and is in contact with the transparent conductive layer **170** to form an ohmic contact. The cladding layer **132**, the graded index layer **136**, and the ohmic contact layer **134** have a photonic crystal structure. For example, the photonic crystal layer **130** has multiple through holes **131** located in the cladding layer **132**, the graded index layer **136**, and the ohmic contact layer **134**, and extends from the cladding layer **132** to the ohmic contact layer **134** to form a photonic crystal structure. In this embodiment, the through hole **131** penetrates through the ohmic contact layer **134**, the graded index layer **136**, and the cladding layer **132**. In addition, in one embodiment, the through holes **131** may be arranged in a two-dimensional array in directions parallel to the substrate **160**. In this embodiment, the photonic crystal surface emitting laser unit **100** further includes a current confinement layer **180** disposed between the photonic crystal layer **130** and the transparent conductive layer **170**, and has an opening **182**. The transparent conductive layer **170** is connected to the photonic crystal layer **130** through the opening **182**.

(25) In this embodiment, the photonic crystal surface emitting laser unit **100** further includes a graded index layer **190** disposed between the substrate **160** and the doped semiconductor layer **110**, where the doped semiconductor layer **110** is a cladding layer.

(26) In addition, in FIG. **1B**, a periodic direction **P1** of the diffractive grating **300** is along a line I-I. However, in other embodiments, as shown in FIG. **1D** and FIG. **1E**, the periodic direction **P1** of the diffractive grating **300** may also be along other directions according to the actual required deflection direction of the light beam **122**.

(27) Referring to FIG. **1A** to FIG. **1C** again, the disclosure does not limit the material of each of the film layers. In some embodiments, the material of the substrate **160** may be gallium nitride (GaN), gallium arsenide (GaAs), indium phosphide (InP), gallium antimonide (GaSb), indium arsenide (InAs), or other suitable materials. In other embodiments, when the first electrode **140** may be electrically connected to the doped semiconductor layer **110** without being electrically connected to the doped semiconductor layer **110** through the substrate **160**, the substrate **160** may be a semi-insulating substrate or an insulating substrate, such as a sapphire substrate. The material of the light-emitting layer **120** may include gallium nitride, aluminum gallium nitride (AlGaIn), gallium arsenide, indium gallium arsenide (InGaAs), aluminum gallium arsenide (AlGaAs), gallium phosphide (GaP), indium arsenide (InAs), indium arsenic antimonide (InAsSb), indium gallium arsenic antimonide (InGaAsSb), aluminum gallium arsenic antimonide (AlGaAsSb), indium gallium arsenide phosphide (InGaAsP), aluminum indium gallium arsenide (AlInGaAs), other suitable semiconductor materials, or a combination thereof, and the light-emitting layer **120** may adopt two or more different materials, or two or more materials of the same compound but with different elemental ratios to form a quantum well layer, a multiple quantum well layer, or a quantum dot structure. The materials of the cladding layer **132**, the doped semiconductor layer **110**, and the graded index layer **136** and the graded index layer **190** are, for example, aluminum gallium arsenide (AlGaAs), gallium arsenide, aluminum gallium nitride, aluminum gallium indium arsenide (AlGaInAs), aluminum gallium indium phosphide (AlGaInP), aluminum gallium arsenide antimonide, or other suitable materials. The material of the ohmic contact layer **134** may be gallium nitride, gallium arsenide, indium gallium arsenide phosphide (InGaAsP), indium gallium arsenide (InGaAs), or other suitable materials. In addition, the ohmic contact layer **134** may be heavily doped with beryllium, carbon, zinc, or a combination thereof, to form P-type doping, so as to establish a good ohmic contact with the transparent conductive layer **170**. The concentration of the beryllium doping may be about 10^{19} cm.⁻³, the concentration of the carbon doping may be about 10^{19} cm.⁻³ to 10^{20} cm.⁻³, and the concentration of the zinc doping may be about 10^{19} cm.⁻³ to 10^{20} cm.⁻³, but the disclosure is not limited thereto. The material of the current confinement layer **180** may be silicon nitride, silicon oxide, or other suitable materials. The current confinement layer **180** may block current, so that the current is

concentrated at the opening **182** and passes through the opening **182**.

(28) The material of the transparent conductive layer **170** is, for example, indium tin oxide (ITO), antimony tin oxide (ATO), fluorine doped tin oxide (FTO), aluminum zinc oxide (AZO), gallium zinc oxide (GZO), indium zinc oxide (IZO), zinc oxide (ZnO), graphene, or other suitable transparent conductive materials. The first electrode **140** and the second electrode **150** may be metal electrodes, the material thereof is, for example, gold, titanium-gold alloy, titanium-platinum-gold alloy, nickel-germanium-gold alloy, or other suitable metals. The transparent conductive layer **170** may conduct current to concentrate the current to pass through the opening **182**, and at the same time, the transparent conductive layer **170** may also allow the light beam **122** emitted by the light-emitting layer **120** to penetrate without blocking the light beam **122**.

(29) In addition, in one embodiment, in each of the above-mentioned film layers, the P-type doping may be beryllium, carbon, zinc, or a combination thereof, and the doping concentration thereof is, for example, about 10^{17} cm⁻³ to 10^{18} cm⁻³ (Except that the doping concentration of the ohmic contact layer **134** is about 10^{19} cm⁻³ to 10^{20} cm⁻³), and the N-type doping may be silicon doping, and its doping concentration is, for example, about 10^{17} cm⁻³ to 10^{18} cm⁻³, but the disclosure is not limited thereto. In other embodiments, other suitable elements may also be adopted for doping to achieve P-type doping and N-type doping.

(30) In this embodiment, the substrate **160**, the graded index layer **190**, and the doped semiconductor layer **110** are all N-type doped semiconductor layers, while the photonic crystal layer **130** including the cladding layer **132**, the graded index layer **136**, and the ohmic contact layer **134** are all P-type doped semiconductor layers. However, in other embodiments, the substrate **160**, the graded index layer **190**, and the doped semiconductor layer **110** may all be P-type doped semiconductor layers, while the cladding layer **132**, the graded index layer **136**, and the ohmic contact layer **134** are all N-type doped semiconductor layers.

(31) The material of the diffractive grating **300** may be a metal, silicon, a semiconductor, a transparent conductive layer (such as indium tin oxide, indium zinc oxide, or aluminum zinc oxide, etc.). The diffractive grating **300** may be a one-dimensional grating (such as a stripe-shaped grating) or a two-dimensional grating (such as a line array, a through hole array, a column array, or a polygonal array), or a microlens array.

(32) FIG. 2A is a cross-sectional schematic view of a photonic crystal surface emitting laser device of another embodiment of the disclosure, FIG. 2B is a simplified perspective schematic view of a photonic crystal layer, a diffractive grating surface of a transparent conductive layer, and a diffractive grating in FIG. 2A. Referring to FIG. 2A and FIG. 2B, a photonic crystal surface emitting laser device **200a** of this embodiment is similar to the photonic crystal surface emitting laser device **200** of FIG. 1C, and the differences between the two are as described below. In a photonic crystal surface emitting laser unit **100a** of the photonic crystal surface emitting laser device **200a**, a surface of a transparent conductive layer **170a** facing away from the light-emitting layer **120** is a diffractive grating surface **172a**. A periodic direction of the diffractive grating surface **172a** and the periodic direction of the diffractive grating **300** may be the same or different, and FIG. 2B is illustrated with an acute angle between the periodic direction of the diffractive grating **300** and the periodic direction of the diffractive grating surface **172a** as an example. When the light beam **122** is diffracted twice by the diffractive grating surface **172a** and the diffractive grating **300**, more changes may occur to the deflection angle. For example, as shown in FIG. 2B, the light beam **122** may be divided into two beams after being diffracted by the diffractive grating surface **172a**, and the two light beams **122** may be divided into a total of four light beams **122** after being diffracted by the diffractive grating **300**. An included angle of the divided light beams **122** may be controlled by a period of the diffractive grating. On the other hand, in the photonic crystal surface emitting laser device **200** shown in FIG. 1C, since the surface of the transparent conductive layer **170** facing away from the light-emitting layer **120** is a smooth surface, the light beam **122** is

divided into two light beams **122** only by being diffracted by the diffractive grating **300**.

(33) In this embodiment, the diffractive grating surface **172a** may be formed naturally by a deposition process of the transparent conductive layer **170a** when a thicker transparent conductive layer **170a** is deposited, or may be additionally formed by an etching process.

(34) In one embodiment, the diffractive grating surface **172a** may be a one-dimensional grating (such as a stripe-shaped grating) or a two-dimensional grating (such as a line array, a through hole array, a column array, or a polygonal array), or a microlens array.

(35) FIG. 3 is a cross-sectional schematic view of a photonic crystal surface emitting laser device of yet another embodiment of the disclosure. Referring to FIG. 3, a photonic crystal surface emitting laser device **200c** of this embodiment is similar to the photonic crystal surface emitting laser device **200** of FIG. 1C, and the differences between the two are as described below. A photonic crystal surface emitting laser unit **100c** of the photonic crystal surface emitting laser device **200c** of this embodiment further includes a distributed Bragg reflector (DBR) **195** disposed between the doped semiconductor layer **110** and the substrate **160**, so that the light beam **122** diffracted from the photonic crystal layer **130** toward the bottom of FIG. 3 is reflected in the direction of the diffractive grating **300**. In this way, the light energy of the light beam **122** exiting from the diffractive grating **300** may be increased, so as to achieve better light utilization efficiency.

(36) FIG. 4 is a cross-sectional schematic view of a photonic crystal surface emitting laser device of another embodiment of the disclosure. Referring to FIG. 4, a photonic crystal surface emitting laser device **200d** of this embodiment is similar to the photonic crystal surface emitting laser device **200** of FIG. 1C, and the differences between the two are as described below. In a photonic crystal surface emitting laser unit **100d** of the photonic crystal surface emitting laser device **200d** of this embodiment, the doped semiconductor layer **110** and the first electrode **140** are disposed on the same side of the substrate **160**.

(37) FIG. 5 is a cross-sectional schematic view of a photonic crystal surface emitting laser device of yet another embodiment of the disclosure. Referring to FIG. 5, a photonic crystal surface emitting laser device **200e** of this embodiment is similar to the photonic crystal surface emitting laser device **200** of FIG. 1C, and the differences between the two are as described below. In a photonic crystal surface emitting laser unit **100e** of the photonic crystal surface emitting laser device **200e** of this embodiment, the diffractive grating **300** is disposed on one side of the doped semiconductor layer **110**, and in the example of FIG. 5, the substrate **160** is disposed between the doped semiconductor layer **110** and the diffractive grating **300**. In this embodiment, the diffractive grating **300** may be a portion of the first electrode **140**, that is, periodically arranged elongated slits **142** are at a center of the first electrode **140** to form the diffractive grating **300**. Alternatively, in other embodiments, the material of the diffractive grating **300** may also be other metals, semiconductors, or oxides different from the material of the first electrode **140**. The first electrode **140** has an opening in the center, and the diffractive grating **300** is disposed in the opening.

(38) On the other hand, in this embodiment, the photonic crystal layer **130** is covered by a second electrode **150e**, and the light beam **122** exits toward the bottom of FIG. 5.

(39) FIG. 6 is a cross-sectional schematic view of a photonic crystal surface emitting laser device of yet another embodiment of the disclosure. Referring to FIG. 6, a photonic crystal surface emitting laser device **200f** of this embodiment is similar to the photonic crystal surface emitting laser device **200e** of FIG. 5, and the differences between the two are as described below. In a photonic crystal surface emitting laser unit **100f** of the photonic crystal surface emitting laser device **200f** of this embodiment, the doped semiconductor layer **110** and the first electrode **140** are disposed on the same side of the substrate **160**, and the diffractive grating **300** and the first electrodes **140** are respectively disposed on opposite sides of the substrate **160**.

(40) FIG. 7 is a cross-sectional schematic view of a photonic crystal surface emitting laser device of yet another embodiment of the disclosure. Referring to FIG. 7, a photonic crystal surface

emitting laser device **200g** of this embodiment is similar to the photonic crystal surface emitting laser device **200e** of FIG. 5, and the differences between the two are as described below. In a photonic crystal surface emitting laser unit **100g** of the photonic crystal surface emitting laser device **200g** of this embodiment, the surface of the substrate **160g** facing away from the light-emitting layer **120** has a diffractive grating surface structure **162g** to form a diffractive grating **300g**.

(41) FIG. 8 is a cross-sectional schematic view of a photonic crystal surface emitting laser device of another embodiment of the disclosure. Referring to FIG. 8, a photonic crystal surface emitting laser device **200h** of this embodiment is similar to the photonic crystal surface emitting laser device **200f** of FIG. 6, and the differences between the two are as described below. In a photonic crystal surface emitting laser unit **100h** of the photonic crystal surface emitting laser device **200h** of this embodiment, the surface of the substrate **160h** facing away from the light-emitting layer **120** has a diffractive grating surface structure **162h** to form a diffractive grating **300h**.

(42) FIG. 9 is a cross-sectional schematic view of a photonic crystal surface emitting laser device of yet another embodiment of the disclosure. Referring to FIG. 9, a photonic crystal surface emitting laser device **200j** of this embodiment is similar to the photonic crystal surface emitting laser device **200** of FIG. 1C, and the differences between the two are as described below. In a photonic crystal surface emitting laser unit **100j** of the photonic crystal surface emitting laser device **200j** of this embodiment, the cladding layer **132j** has a photonic crystal structure, and is covered by a semiconductor layer **132'**. The material of the semiconductor layer **132'** is similar to or the same as that of the cladding layer **132j**, and may be a P-type or an N-type doped semiconductor layer along with the cladding layer **132j**. A graded index layer **136j** and an ohmic contact layer **134j** may be sequentially disposed on the semiconductor layer **132'**. The edges of the semiconductor layer **132'** and the ohmic contact layer **134j** are covered by the current confinement layer **180**, and the center of the ohmic contact layer **134j** is exposed by the opening **182** of the current confinement layer **180**. The second electrode **150** is connected to the ohmic contact layer **134j** through the opening **182**, and the diffractive grating **300** is located in the center of the ohmic contact layer **134j**.

(43) FIG. 10 is a cross-sectional schematic view of a photonic crystal surface emitting laser device of yet another embodiment of the disclosure. Referring to FIG. 10, a photonic crystal surface emitting laser device **200k** of this embodiment is similar to the photonic crystal surface emitting laser device **200e** of FIG. 5 and the photonic crystal surface emitting laser device **200j** of FIG. 9, and the differences are as described below. In a photonic crystal surface emitting laser unit **100k** of the photonic crystal surface emitting laser device **200k** of this embodiment, the photonic crystal layer **130** of FIG. 5 are replaced by the cladding layer **132j**, the semiconductor layer **132'**, the graded index layer **136j**, and the ohmic contact layer **134j** of FIG. 9, and the photonic crystal surface emitting laser unit **100k** does not include the transparent conductive layer **170**. The second electrode **150k** is connected to the ohmic contact layer **134j** through the opening **182**.

(44) FIG. 11 is a cross-sectional schematic view of a photonic crystal surface emitting laser device of yet another embodiment of the disclosure. Referring to FIG. 11, a photonic crystal surface emitting laser device **200m** of this embodiment is similar to the photonic crystal surface emitting laser device **200k** of FIG. 10, and the differences between the two are as described below. A photonic crystal surface emitting laser unit **100m** of the photonic crystal surface emitting laser device **200m** of this embodiment further includes a distributed Bragg reflector **195** disposed between the semiconductor layer **132'** and the graded index layer **136j**.

(45) In other embodiments, the photonic crystal layer **130** in FIG. 6, FIG. 7, and FIG. 8 and the structures thereon may also be changed to a cladding layer **132j**, a semiconductor layer **132'**, a graded index layer **136j**, an ohmic contact layer **134j**, and a second electrode **150k**, as shown in FIG. 10 or FIG. 11, and do not have a transparent conductive layer **170**. That is, the structures above the light-emitting layer **120** in FIG. 6, FIG. 7, and FIG. 8 may be replaced with, for example,

the structures above the light-emitting layer **120** in FIG. **10** or FIG. **11**, to form other six types of photonic crystal surface emitting laser units.

(46) FIG. **12** is a cross-sectional schematic view of a photonic crystal surface emitting laser device of yet another embodiment of the disclosure. Referring to FIG. **12**, a photonic crystal surface emitting laser device **200i** of this embodiment is similar to the photonic crystal surface emitting laser device **200** of FIG. **1A**, and the differences between the two are as described below. The photonic crystal surface emitting laser device **200i** of this embodiment includes multiple photonic crystal surface emitting laser units **100** arranged in an array. FIG. **12** is illustrated with 2×2 photonic crystal surface emitting laser units **100** as an example. However, in other embodiments, the array may be $M \times N$ photonic crystal surface emitting laser units **100**, where M and N are any positive integers. For example, in one embodiment, the photonic crystal surface emitting laser device **200i** may include 10×10 photonic crystal surface emitting laser units **100**. At least one of the periods and the arrangement directions of multiple diffractive gratings **300** of the photonic crystal surface emitting laser units **100** are different. For example, a diffractive grating **3001**, a diffractive grating **3002**, a diffractive grating **3003**, and a diffractive grating **3004** in FIG. **12** have different periods and/or arrangement directions (i.e., periodic directions). For example, different periodic directions **P1** as shown in FIG. **1B** to FIG. **1E** may be adopted to form different exiting directions of the light beams **122**.

(47) In this embodiment, the photonic crystal surface emitting laser units **100** are electrically connected to a controller **210**. The controller **210** may individually and independently control the emitted light of the photonic crystal surface emitting laser units **100**. For example, since the light-emitting directions of the photonic crystal surface emitting laser units **100** are different, the controller **210** may control the photonic crystal surface emitting laser units **100** to sequentially emit light beams **122** (i.e., laser beams) in different directions, to achieve beam scanning. Since such beam scanning is electronically controlled rather than mechanically controlled, this embodiment attains faster scanning speeds, higher reliability, and better durability. Such a photonic crystal surface emitting laser device **200i** may be applied to light detection and distance measurement, such as being applied to lidar or three-dimensional sensing, and has the advantages of high speed, high reliability, and high durability. In addition, this kind of laser that does not rely on mechanical actuation, has high power, has high beam quality and may achieve two-dimensional scanning (i.e., the photonic crystal surface emitting laser device **200i**) may also be applied to the sensing system of smart mobility, object recognition system, or adaptive lighting, etc.

(48) In this embodiment, the controller **210** may be designed through a hardware description language (HDL) or any other digital circuit and analog circuit design method known to those with ordinary knowledge in the art, and the hardware circuit may be realized through a field programmable gate array (FPGA), a complex programmable logic device (CPLD), or an application-specific integrated circuit (ASIC).

(49) In addition, the photonic crystal surface emitting laser unit **100** in the photonic crystal surface emitting laser device **200i** may also be replaced with the photonic crystal surface emitting laser units of other embodiments described above (for example, the photonic crystal surface emitting laser unit **100a** to the photonic crystal surface emitting laser unit **100h**) to form various photonic crystal surface emitting laser devices having photonic crystal surface emitting laser units arranged in an array.

(50) To sum up, in the photonic crystal surface emitting laser device of the embodiment of the disclosure, since the diffractive grating disposed on the photonic crystal layer or the doped semiconductor layer is adopted to diffract the laser beam diffracted by the photonic crystal layer, therefore the exiting light deflection direction of the laser beam may be determined by using a simple structure. The photonic crystal surface emitting laser device of an embodiment of the disclosure may include multiple photonic crystal surface emitting laser units arranged in an array, and at least one of the periods and the arrangement directions of multiple diffractive gratings **300** of

the photonic crystal surface emitting laser units **100** are different. Since the light-emitting directions of the photonic crystal surface emitting laser units are different, the controller may control the photonic crystal surface emitting laser units to sequentially emit laser beams in different directions, to achieve beam scanning. Since such beam scanning is electronically controlled rather than mechanically controlled, the embodiment attains faster scanning speeds, higher reliability, and better durability. Such a photonic crystal surface emitting laser device may be applied to light detection and distance measurement, such as being applied to lidar or three-dimensional sensing, and has the advantages of high speed, high reliability, and high durability.

Claims

1. A photonic crystal surface emitting laser device, comprising: at least one photonic crystal surface emitting laser unit, the photonic crystal surface emitting laser unit comprising: a light-emitting layer, configured to emit a light beam; a photonic crystal layer, disposed on one side of the light-emitting layer; a doped semiconductor layer, disposed on another side of the light-emitting layer; and a diffractive grating, disposed on the photonic crystal layer or the doped semiconductor layer; and a first electrode, electrically connected to the doped semiconductor layer, wherein the photonic crystal surface emitting laser unit further comprises a second electrode, disposed on the photonic crystal layer and electrically connected to the photonic crystal layer, and wherein a portion of the second electrode is the diffractive grating, and after the light beam is diffracted by the photonic crystal layer and the diffractive grating, the light beam is emitted from the photonic crystal surface emitting laser unit.
 2. The photonic crystal surface emitting laser device according to claim 1, wherein the photonic crystal surface emitting laser unit further comprises a transparent conductive layer disposed between the photonic crystal layer and the second electrode.
 3. The photonic crystal surface emitting laser device according to claim 1, further comprising a substrate, disposed between the doped semiconductor layer and the first electrode.
 4. The photonic crystal surface emitting laser device according to claim 1, further comprising a substrate, wherein the doped semiconductor layer and the first electrode are disposed on a same side of the substrate.
 5. The photonic crystal surface emitting laser device according to claim 1, further comprising a substrate, disposed between the doped semiconductor layer and the diffractive grating.
 6. The photonic crystal surface emitting laser device according to claim 1, wherein the at least one photonic crystal surface emitting laser unit is a plurality of photonic crystal surface emitting laser units arranged in an array, at least one of periods and arrangement directions of a plurality of diffractive gratings of the photonic crystal surface emitting laser units are different.
-